

MATH 4162/MATH 6201

Numerical methods for evolutionary differential equation

Fall 2025

COURSE PROFESSOR

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Access your course in <https://online.mun.ca/>

MEETING

The class is scheduled on TR from 09:00 to 10:15 in SN-3042.

OFFICE HOUR

Office hour: (T)R 1500-1600 in HH3054

Calendar description

This course covers numerical solution of initial value problems for ordinary differential equations by single and multi-step methods, Runge-Kutta, and predictor-corrector; numerical solution of boundary value problems for ordinary differential equations by shooting methods, finite differences and spectral methods; numerical solution of partial differential equations by the method of lines, finite differences, finite volumes and finite elements.

COURSE OUTLINE

We will start from stability of differential equations and approximation of derivatives. Then, we will move toward numerical methods covering finite difference methods, spectral methods, finite volume methods, and finite element methods. Additional topics arising from the discretization and solution of specific equations will be covered, i.e. stability, consistency, convergence, iterative methods, solvers by Fast Fourier Transforms (FFT), and implementation in python, matlab, and/or C++ programming.

Tentative lecture plan:

1. Discretization and stability of PDEs (2 weeks)
2. Finite Difference (FD) Methods (4 weeks)
3. Finite Volume (FV) Methods (2 weeks)
4. Spectral method (2 weeks)
5. Finite Element Methods (FEM) (2 weeks)

Learning Objectives

- i) Traditional numerical solvers for modelling real world problems with examples drawing from energy and environment;
- ii) Concept of modern data-driven numerical solvers for simulating complex systems;
- iii) Utilization of data-science techniques for scientific analysis and presentation of modelling method.

Generative Artificial Intelligence (GAI)

The submission of work that has been created by generative artificial intelligence (GAI) tools and presented as a student's original work is considered an academic offence in this course, and your work will be subject to the appropriate Memorial's Academic Misconduct policy.

MARKING SCHEME

Assignments and case studies	25%	
Projects	30%	
Assignment 1	5%	Sep 26, 2025 at 1700
Case study 1	5%	Oct 10, 2025 at 1700
Midterm report	10%	Oct 17, 2025 at 1200
Assignment 2	5%	Oct 24, 2025 at 1700
Case study 2	5%	Nov 07, 2025 at 1700
Assignment 3	5%	Nov 21, 2025 at 1700
Final report	20%	Dec 04, 2025 at 2359
Final exam	45%	see exam schedule

PROJECT

The final report should have the format (and professional look) of a journal article. It should cover background of the problem chosen, discretization of governing equation and stability analysis, results and conclusion with future direction.

The project may be related to students' own research interests, but cannot be recycling of previous projects. The results do not have to be novel or original, but must be a sound investigation of an actual topic.

Students are strongly encouraged to consider a simulation or digital twin of a phenomena and focus on utilizing data-science, machine learning techniques, which may utilize neural networks (although not covered in the course). The goal is to study a relevant topic, which applies the methods covered in this course.

The midterm report will show details for the derivation of governing equations, discretization, and stability analysis. The final report will briefly include points covered in midterm along with other points mentioned above, such as result analysis and findings.

PREREQUISITE

MATH 2260, MATH 3132 and MATH 3202.

MATH 2260 is essential to gain knowledge of solution of linear differential equations and systems. Students with equivalent knowledge in computational techniques and calculus may take the course if they have not completed 3132/3202, subject to permission of course instructor.

TEXT BOOK

There will be no fixed textbook, but recommended books are listed below. Here, '*' indicates the books were primarily considered in organizing lecture notes.

- *Arieh Iserles, A First Course in the Numerical Analysis of Differential Equations
- Adler, James and Sterck, Hans De and Maclachlan, Scott and Olson, Luke, Numerical partial differential equations.
- *Finite difference methods for ordinary differential and partial differential equations by Randall LeVeque.
- Spectral methods in Matlab by Lloyd Trefethen (available in the pdf format).
- K. W. Morton and D. F. Mayers, Numerical Solution of Partial Differential Equations, Cambridge,
- Lecture notes by J. Alam (available only during lectures of math 4162).
- Any other text book of your choice.

Plagiarism and other academic misconduct:

Any instance of plagiarism or cheating or any other form of academic misconduct will be treated according to Section 6.12 of the University Calendar. Students are advised to consult the relevant sections for full details.

Note that section 6.12.4 includes the following statements of **Academic offences**:

- **Impersonating** another student or allowing oneself to be impersonated: This includes the imitation of a student or the entrance into an arrangement with another person to be impersonated for the purposes of taking examinations or tests or carrying out laboratory or other assignments.
- **Plagiarism**: Plagiarism is the act of presenting the ideas or works of another as one's own. This applies to all material such as essays, laboratory assignments, laboratory reports, work term reports, design projects, seminar presentations, statistical data, computer programs, research results and theses. The properly acknowledged use of sources is an accepted and important part of scholarship. Use of such material without acknowledgement is contrary to accepted norms of academic behaviour. Information regarding acceptable writing practices is available through the Writing Centre at www.mun.ca/writingcentre

Vector calculus and differential equation: big picture.

Consider a two-dimensional vector field

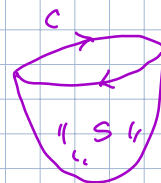
$$\vec{u} = \begin{bmatrix} u(x, y, t) \\ v(x, y, t) \end{bmatrix}$$

If a particle moving along a path c in this vector field, then the work done by this vector field is $\int_c \vec{u} \cdot d\vec{r}$

If the vector field is conservative, then the path integral is independent of the choice of the path.

Proof: By Stokes theorem

$$\oint_c \vec{u} \cdot d\vec{r} = \iint_S \nabla \times \vec{u} \cdot \hat{n} \, ds$$



Here $\nabla = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle$ is called gradient operator.

Curl is defined as $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$

F_x, F_y, F_z are components of the vector $\vec{F}$. (or $\vec{u}$)

For our 2D vector field $\nabla \times \vec{u} = \hat{k} \left[\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right]$

Since $\vec{u}$ is conservative, we have $\nabla \times \vec{u} = 0$

$\oint_c \vec{u} \cdot d\vec{r} = 0$ Hence $\int_c \vec{u} \cdot d\vec{r}$ is independent of path.

If we drop a particle at position $\vec{x} = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$ at $t=0$ in this vector field, the path of the particle is given by

$$\frac{d\vec{x}}{dt} = \vec{u}(x, t)$$

Which is a system of ordinary differential equation

The vector field itself is governed by a partial differential equation

$$\frac{\partial \vec{u}}{\partial t} = N(\vec{u})$$

where N a partial differential operator.

The divergence of the 2D vector field is

$$\nabla \cdot \vec{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}$$

The divergence of the gradient of a component of $\vec{u}$ is

$$\nabla \cdot \nabla u = \nabla \cdot \begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \end{bmatrix} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

H.W. Review Stoke's theorem and divergence theorem

Classification of partial differential equations

Second order linear PDE

$$\underbrace{Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu}_{\text{principal part}} = G$$

non-homogeneous

Non linear PDE will have nonlinear terms.

The characteristic equation is

$$\frac{dy}{dx} = \frac{1}{2A} (B \pm \sqrt{B^2 - 4AC})$$

Three cases:

I) $B^2 - 4AC > 0$, PDE is hyperbolic

$$\text{Ex: } \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$$

$$A=1, B=0, C=-a^2$$

$$B^2 - 4AC = 0^2 - 4(1)(-a^2) = 4a^2 > 0$$

There are two characteristic curves

ii) $B^2 - 4AC = 0$, PDE is parabolic

$$\text{Ex: } \frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial x^2}$$

$$A = 0, B = 0, C = -\nu$$

$$B^2 - 4AC = 0 - 4(0)(\nu) = 0$$

PDE has a single characteristic curve.

iii) $B^2 - 4AC < 0$, PDE is elliptic.

$$\text{Ex: } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \sin x \cos y$$

$$A = 1, B = 0, C = 1$$

$$B^2 - 4AC = 0 - 4(1)(1) = -4 < 0$$

No real characteristic curves

Ex: Classify the PDEs and find characteristic equations

$$u_{xx} + 5u_{xy} + 6u_{yy} = 0$$

$$y^2 u_{xx} - 2y u_{xy} + u_{yy} = u_x + 6y$$

A model partial differential equation

$$\boxed{\frac{\partial u}{\partial t} = Lu + q}, \quad x \in \Omega \subset \mathbb{R}^n, \quad n=1,2,3$$

L is a differential operator

Example:

$$L = -a \frac{\partial}{\partial x}, \quad q = 0$$

$$\frac{\partial u}{\partial t} = -a \frac{\partial u}{\partial x}$$

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0 \quad \Rightarrow \text{advection.}$$

$$\text{let } u(x,t) = f(x-at)$$

$$\frac{\partial u}{\partial t} = -a f', \quad \frac{\partial u}{\partial x} = f'$$

Then

$$-a f' + a f' = 0$$

Hence $f(x-at)$ is a solution.

$$\text{At } t=0, \quad u(x,0) = f(x)$$

This is called initial condition.

At any later time, $u(x,t)$ is $f(x)$ shifted by ct unit.

We can say that a wave propagates at a speed of c

Diffusion.

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial x^2}, \quad \nu > 0$$

$$\text{let } u(x, t) = e^{-\nu t} \sin(x)$$

$$\frac{\partial u}{\partial t} = -\nu e^{-\nu t} \sin(x)$$

$$\frac{\partial u}{\partial x} = e^{-\nu t} \cos x$$

$$\frac{\partial^2 u}{\partial x^2} = -e^{-\nu t} \sin x$$

$$\nu \frac{\partial^2 u}{\partial x^2} = -\nu e^{-\nu t} \sin x$$

Hence $u(x, t) = e^{-\nu t} \sin x$ is a solution

Initial condition is

$$u(x, 0) = \sin x$$

Remark:

Advection equation is hyperbolic

Diffusion " " parabolic

Laplace equation is elliptic

Aim be able to identify PDEs as hyperbolic, parabolic, elliptic, linear, nonlinear etc.

Solution of differential equation

let $\frac{du}{dt} = au$, a is a scalar

$$u(t) = u_0 e^{at}, \quad u(0) = u_0 \text{ is initial condition}$$

let $\frac{d\vec{u}}{dt} = A\vec{u}$, A is $n \times n$ square matrix
 $\vec{u}$ is $n \times 1$ vector

$$\vec{u}(t) = \vec{u}_0 e^{At}$$

let $A = \phi \Lambda \phi^{-1}$ be the eigen decomposition

$$\vec{u}(t) = \phi e^{\Lambda t} \phi^{-1} u_0, \quad \Lambda = \text{diag}(\lambda_i), i=1, \dots, n$$

Using the concept of 1st order ODE and eigen decomposition, we solve the system of linear ODEs. Any system of non-linear ODE or PDE can be converted to a linear system before solving it.

let $\frac{\partial u}{\partial t} = Lu$, L is a linear differential operator.

$u(x,t) = u_0 e^{\lambda t}$, Where $Lu = \lambda u$, λ is eigenvalue
 $u_0(x)$ is to be found

let λ_k be an eigenvalue

$$L\phi_k = \lambda_k \phi_k$$

ϕ_k is corresponding eigenfunction.

We see that $u_k = c_k e^{\lambda_k t} \phi_k(x)$ is a solution.

In general, $u(x,t) = \sum_k c_k e^{\lambda_k t} \phi_k(x)$

Take home message:

If a linear PDE is a good approximation of a target PDE, then solution of linear PDE will be an approximate solution of target PDE.

By solving a linear system of ODE, we can get approximate solution of any PDE subject to certain condition.

Conservation law

Consider the density $\rho = \rho(x, t, u, u_x, \dots)$

and the flux $F = F(x, t, u, u_x, \dots)$

In general $\vec{u}(\vec{x}, t)$ is a vector field

and flux of a scalar $\rho(\vec{x}, t)$ is

$$\vec{F} = \rho \vec{u}$$

Consider a boundary condition

$$F \rightarrow \text{Constant as } |x| \rightarrow \infty$$

Conservation law states that

$$\frac{d}{dt} \int_{-\infty}^{\infty} \rho \, dx = 0$$

In other words $\int_{-\infty}^{\infty} \rho \, dx = \text{Constant}$

this constant is a conserved quantity

and is usually called a constant of

motion or an invariant of a corresponding

PDE.

Consider the PDE:

$$\frac{\partial \rho}{\partial t} + \frac{\partial F}{\partial x} = 0, \quad F = F(\rho) = \frac{1}{2} \rho^2 \text{ for example.}$$

Integrate
$$\int_{-\infty}^{\infty} \frac{\partial \rho}{\partial t} dx + \int_{-\infty}^{\infty} \frac{\partial F}{\partial x} dx = 0$$

$$\frac{\partial}{\partial t} \int_{-\infty}^{\infty} \rho dx + \lim_{x \rightarrow \pm \infty} |F| = 0$$

$$\frac{\partial}{\partial t} \int_{-\infty}^{\infty} \rho dx = 0$$

H.W.
$$\frac{\partial u}{\partial t} - 6u \frac{\partial u}{\partial x} + \frac{\partial^3 u}{\partial x^3} = 0 \quad (\text{KdV equation})$$

$$u = u(x, t), \quad t \in \mathbb{R}, x \in \mathbb{R}$$

i) show that $\int_{-\infty}^{\infty} u dx = \text{Constant}$ is an

invariant. Find the flux

ii) Multiply the PDE by u and integrate

$$\frac{\partial}{\partial t} \frac{1}{2} u^2 - 6u^2 \frac{\partial u}{\partial x} + u \frac{\partial^3 u}{\partial x^3} = 0$$

$$\frac{\partial}{\partial t} \frac{1}{2} u^2 + \frac{\partial}{\partial x} \left(-2u^3 + u \frac{\partial^2 u}{\partial x^2} - \frac{1}{2} \left(\frac{\partial u}{\partial x} \right)^2 \right) = 0$$

$$\text{flux} = -2u^3 + u u_{xx} - \frac{1}{2} (u_x)^2$$

$$\int_{-\infty}^{\infty} \frac{1}{2} u^2 dx = \text{Constant} \text{ is an invariant.}$$

Lecture #2, Sep 11, 2025

Fourier analysis method

There are two fundamental approaches for proving stability of numerical methods. One is Fourier analysis (von Neumann analysis).

It applies only to linear constant coefficient PDEs.

The other is energy method, which can be applied to nonlinear PDEs with variable coefficients.

Lax-Richtmyer Equivalence theorem:

Given a well-posed linear initial value problem and its consistent numerical approximation, stability is **necessary and sufficient** condition for convergence.

Proving convergence of any numerical method is often difficult. However stability is easier than showing convergence.

spatial Fourier mode.

Fourier transform is defined as:

$$FT\{f\} = \hat{f}(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{ikx} dx \quad (\text{forward transform})$$

e^{ikx} is called Fourier mode

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(k) e^{-ikx} dk \quad (\text{Inverse transform})$$

Here $k = \frac{2\pi}{\lambda}$ is called wavenumber and
 λ is called wavelength.

often $\omega = \frac{2\pi}{T}$ is called angular frequency
for a period of T if Fourier transform
is considered in time domain.

Example: 1)
$$f(x) = \begin{cases} 0 & |x| > L \\ 1 & |x| < L \end{cases}$$

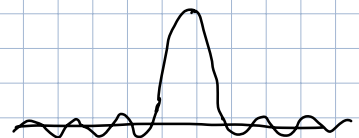


$$\hat{f}(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{ikx} dx$$

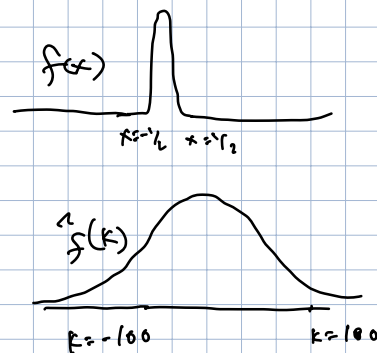
$$\text{sinc}(x) = \frac{\sin x}{x}$$

$$= \frac{1}{\pi} \frac{\sin kL}{k}$$

$$= \frac{L}{\pi} \text{sinc}(kL)$$



$$\begin{aligned}
 2) \quad f(x) &= -e^{-\alpha x^2} \\
 \hat{f}(k) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{ikx} dx \\
 &= \frac{1}{\sqrt{4\pi\alpha}} e^{-\frac{k^2}{4\alpha}}
 \end{aligned}$$



If $f(x)$ has a relatively narrow band in space,
it will require a relatively large number of modes
(degrees of freedom) for a consistent numerical method.

③ $f(x-\beta)$ where $f(x)$ is any well-defined function
and $\beta > 0$

$$\text{let } \hat{f}(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{ikx} dx$$

$$\begin{aligned}
 \text{then } \hat{F}(k) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x-\beta) e^{ikx} dx \\
 &= e^{ik\beta} \hat{f}(k)
 \end{aligned}$$

Prev. lecture $\Rightarrow$

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

$$u(x, 0) = f(x)$$

$$u(x, t) = f(x - at)$$

$$= f(x - \beta)$$

H.W. Show that Fourier modes of solution of $\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$
never decays nor grows for any initial condition, $x \in \mathbb{R}$
If waves travel at constant speed, there is no loss of amplitudes.

4)

Convolution of $f(x)$ and $g(x)$ is defined as

$$\begin{aligned}(f * g)(x) &= \int_{-\infty}^{\infty} f(y) g(x-y) dy \\ &= \int_{-\infty}^{\infty} f(x-y) g(y) dy\end{aligned}$$

Fourier transform of convolution becomes multiplication in Fourier domain.

Numerical case: discrete Fourier transform (FFTW package)

let f and g are two sequences defined over an ordered set $\{x_0, x_1, x_2, \dots, x_N\}$.

let the convolutional kernel g is defined over a finite interval $[-M, M]$.

For every x_n with $n=0, 1, \dots, N$,

$$(f * g)[n] = \sum_{k=-M}^M f[n-k] g[k] \quad \text{subject to appropriate boundary conditions.}$$

5)

Fourier transform of derivative

$$\begin{aligned}
 & \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\partial f}{\partial x} e^{ikx} dx \\
 & \left. \begin{aligned} \text{FT} \left\{ \frac{\partial f}{\partial x} \right\} &= -ik \hat{f} \\ \text{FT} \left\{ \frac{\partial^2 f}{\partial x^2} \right\} &= -(ik)^2 \hat{f} \\ &= -k^2 \hat{f} \\ \text{FT} \left\{ \frac{\partial^3 f}{\partial x^3} \right\} &= +ik^3 \hat{f} \end{aligned} \right\} \\
 &= \frac{1}{2\pi} \left[f(x) e^{ikx} \right]_{-\infty}^{\infty} - \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) (ik) e^{ikx} dx \\
 &= 0 - \frac{ik}{2\pi} \int_{-\infty}^{\infty} f(x) e^{ikx} dx \\
 &= \boxed{-ik \hat{f}(k)}
 \end{aligned}$$

If we have a suitable boundary condition, then Fourier transform of $\frac{\partial f}{\partial x}$ becomes a multiplication in Fourier domain.

Thus

$$\frac{\partial f}{\partial x} = \int_{-\infty}^{\infty} -ik \hat{f}(k) e^{-ikx} dk.$$

Algorithm: Compute derivative.

let $f = [f_0, f_1, \dots, f_N]$

Apply FFT method to compute $\hat{f}$

Multiply by $-ik$ to get $-ik \hat{f}$

Do an inverse FFT to get $\frac{\partial f}{\partial x}$

Summary

1) A finite length convolutional kernel attenuates high frequency modes of a function $f(x)$

2) If $\hat{g}(k) = ik$ is the convolutional kernel in the Fourier domain, the convolution of $g(x)$ and $f(x)$ leads to the derivative of $f(x)$

$$\text{Ex: 1) } \frac{\partial u}{\partial t} = \frac{\partial^3 u}{\partial x^3}$$

Using Fourier transform, show that

$$u(x, t) = \int_{-\infty}^{\infty} \hat{u}(k, 0) e^{-ik(x + k^2 t)} dk$$

$$2) \quad \frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} + b \frac{\partial^3 u}{\partial x^3} = 0$$

Show that

$$u(x, t) = \int_{-\infty}^{\infty} \hat{u}(k, 0) e^{-ik(x + (a - bk^2)t)} dk$$

Observe a particular effect of Fourier transform
for odd vs even derivatives

Ex: D $\frac{\partial u}{\partial t} = \frac{\partial^3 u}{\partial x^3}$

Using Fourier transform, show that

$$u(x,t) = \int_{-\infty}^{\infty} \hat{u}(k,0) e^{i k (x - k^2 t)} dk$$

Solⁿ $FT_x \left\{ \frac{\partial u}{\partial t} \right\} = FT_x \left\{ \frac{\partial^3 u}{\partial x^3} \right\}, \quad FT_x \{ f(x) \} = \hat{f}(k)$

$$\frac{\partial}{\partial t} FT_x \{ u \} = FT_x \left\{ \frac{\partial^3 u}{\partial x^3} \right\}$$

$$\frac{\partial \hat{u}(k,t)}{\partial t} = (-ik)^3 \hat{u}(k,t)$$

$$\boxed{\frac{\partial \hat{u}(k,t)}{\partial t} = i k^3 \hat{u}(k,t)}$$

$$\hat{u}(k,t) = e^{i k^3 t} \hat{u}(k,0)$$

$$\frac{du}{dt} = au$$

$$u(t) = u_0 e^{at}$$

$$u(x,t) = \int_{-\infty}^{\infty} e^{i k^3 t} \hat{u}(k,0) e^{-i k x} dk$$

$$= \int_{-\infty}^{\infty} \hat{u}(k,0) e^{-i k (x - k^2 t)} dk$$

Remark 1) $\hat{u}(k,t)$ depends on eigenvalue $i k^3$

2) What happens to a single Fourier mode $\hat{u}(k,t)$

Stability theorem (well-posedness)

Let the Fourier transform of a PDE is

$$\frac{\partial \hat{u}(k,t)}{\partial t} = A(ik) \hat{u}(k,t)$$

Here $A(ik)$ is the Fourier transform of the differential operator.

We have

$$\hat{u}(k,t) = e^{A(ik)t} \hat{u}(k,0)$$

$$\text{Since } \|\hat{u}(k,t)\| \leq \|e^{A(ik)t}\| \|\hat{u}(k,0)\|$$

The Fourier modes will not grow unboundedly if there are constants k and α such that

$$\|e^{A(ik)t}\| \leq k e^{\alpha t}$$

Then the eigenvalues of $A(ik)$ defines the region of well-posedness.

H.W. 1) show that $\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0, \quad a > 0$

is well-posed.

Hint: $A(ik) = -aik$

2) Using Fourier transform, show that $u(x,t) = f(x-at)$ is the solution of the advection equation.

Dispersion and dissipation

Dispersion refers to the phenomena where waves of different frequencies (or wavenumber) travel at different speeds.

Dissipation refers to the loss of wave amplitude over time, [often due to irreversible energy conversion].

While dispersion and dissipation are important characteristics of the solution of PDE, they are crucial to characterize numerical solution of PDEs. Numerical schemes can artificially cause dissipation and dispersion.

To identify artificial dispersion and dissipation, Fourier analysis and (modified) PDE analysis are crucial.

Dispersion relation:

let $k \in \mathbb{R}$ and ω is a complex number.

Often e^{ikx} and $e^{i\omega t}$ are called spatial and temporal Fourier modes.

Following the concept from linear ODEs,

$$u(x, t) = e^{i(kx + \omega t)}$$

may be considered a trial solution of a PDE.

upon substituting into a linear PDE,

we must have a relationship between k and ω

such as

$$\omega = \omega(k)$$

this is called dispersion relation.

Ex: 1) $\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x}$, $\omega(k) = k$,

2) $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, $\omega(k) = i k^2$ or $i \omega(k) = -k^2$

Hint: Dispersion relation can be obtained for discretized PDE

$$1) \frac{\partial u}{\partial t} = e^{i(kx + \omega t)} \frac{d}{dt} (e^{i(kx + \omega t)}) = e^{i(kx + \omega t)} (i\omega)$$

$$\frac{\partial u}{\partial x} = e^{i(kx+\omega t)} \frac{\partial}{\partial x} (i(kx+\omega t)) = e^{i(kx+\omega t)} (ik)$$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x} \Rightarrow e^{i(kx+\omega t)} i\omega = e^{i(kx+\omega t)} i k$$

$$\boxed{\omega = k} \Rightarrow \omega = \omega(k) \equiv k$$

For a linear constant coefficient PDE,

The phase speed is

$$c_p(k) = -\frac{\omega(k)}{k}$$

and group speed is

$$c_g(k) = -\frac{\partial \omega}{\partial k}$$

If the phase speed $c_p(k)$ is a function of k , then the solution is dispersive.

Different frequencies travel at different speeds

If $\text{Im } \omega > 0$, then

$$\begin{aligned} |u(x,t)| &\leq |\hat{u}_0 e^{i(kx + \omega t)}| \\ &= |\hat{u}_0 e^{-t \text{Im } \omega(k)}| \end{aligned}$$

shows that the solution decays exponentially.

the solution is dissipative.

Each Fourier mode decays exponentially.

ent

Example: Consider the scalar advection equation

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0, \quad x \in \mathbb{R}, \quad t > 0$$

$$u(x, 0) = f(x)$$

differential operator is $L = c \frac{\partial}{\partial x}$

$$\hat{L}(ik) = \frac{1}{2\pi} \int_{-\infty}^{\infty} c \frac{\partial u}{\partial x} e^{ikx} dx$$

$$= -cik \hat{u}(k, t)$$

Applying Fourier transform on to the advection equation, we get

$$\frac{\partial \hat{u}(k, t)}{\partial t} - ick \hat{u}(k, t) = 0$$

$$\hat{u}(k, t) = e^{ickt} \hat{u}(k, 0)$$

$$\text{where } \hat{u}(k, 0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{ikx} dx$$

$$\|\hat{u}(k, t)\| \leq \|e^{ickt}\| \|\hat{u}(k, 0)\|$$

Summary of week 1

- Classification of PDE
- Solution of linear system of ODEs
- Conservation laws for PDEs
- Dispersion and dissipation by Fourier Transform.
- Condition for well-posed PDE